

EM-Based Joint DOA and Delay–Doppler Estimation for Passive Coherent Localization

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Abstract—This paper investigates a **Passive Coherent Localization (PCL)** scenario where an array receiver collects a superposition of a direct-path signal and its target-reflected replica. The goal is to jointly estimate the directions of arrival (DOAs) of the two components, as well as the relative time delay and Doppler shift between them. We develop a deterministic signal model that exploits the intrinsic coupling between the direct signal and its target-reflected replica. Based on this model, we formulate a maximum-likelihood (ML) estimation problem and propose an efficient expectation–maximization (EM) algorithm. At each iteration, the E-step decomposes the array observations into source-specific components, while the M-step updates the DOAs via conventional beamforming, reconstructs the source waveforms by least squares, and estimates the delay–Doppler parameters using the cross-ambiguity function (CAF). Simulation results show that the proposed EM-based joint method improves DOA estimation accuracy compared with a sequential baseline, while achieving comparable performance in delay and Doppler estimation, thereby demonstrating the benefit of leveraging the underlying signal-structure coupling for PCL parameter estimation.

Index Terms—Passive Coherent Location, DOA estimation, Time delay, Doppler shift, Prior knowledge.

I. INTRODUCTION

Target localization plays a key role in numerous applications such as autonomous navigation and defense systems [1], [2]. Passive Coherent Localization (PCL), which utilizes signals from existing sources (illuminators of opportunity), has gained significant attention due to its cost-effectiveness and stealth capabilities [3]–[5]. In PCL, target localization typically involves two main steps: first, estimating key parameters like time delay, Doppler shift, and direction of arrival (DOA); second, using these estimates to compute the target's position [6], [7].

While most existing studies on PCL focus on the second step, the parameter estimation step is equally crucial and often overlooked in the context of PCL. Accurate estimation of these parameters is essential for improving the overall localization performance [8], [9]. Previous research has explored joint estimation of time delay and Doppler shift, particularly for passive systems with non-cooperative transmitters [6]. While

many PCL studies assume separate reference and surveillance channel signals, practical scenarios may involve signal leakage between channels, resulting in direct-path interference (DPI). This paper considers such a scenario where both direct and reflected signals are superimposed in one channel. Rather than using a single output receiver, this work utilizes a multi-output sensor array receiver to capture the composite signal.

Sensor arrays provide spatial diversity [10], enhancing the extraction of time delay, Doppler shift, and DOA. Many algorithms for DOA estimation have been proposed, including beamforming-based methods [11], subspace methods [12], and sparse methods [13]. In PCL, prior knowledge about the signals (i.e., delayed and Doppler-shifted versions of the same signal) can improve estimation accuracy. However, limited attention has been paid to prior waveform knowledge in PCL scenarios.

This paper addresses the joint estimation of DOA and delay–Doppler using a maximum likelihood (ML) approach, employing the expectation–maximization (EM) algorithm for efficient computation [14], [15]. The proposed method leverages a deterministic signal model to account for the relationship between the direct and reflected signals, enhancing the estimation accuracy [16], [17]. The use of prior waveform knowledge, such as the delay and Doppler-shifted characteristics of the signals, allows for better performance than traditional DOA estimation methods [18], [19]. The EM algorithm is applied to iteratively solve the ML estimation problem, ensuring optimal parameter estimation despite the complexity introduced by signal superposition in PCL scenarios.

II. SIGNAL MODEL

We consider a passive sensing scenario in which a receive array observes signals emitted by a single transmitter through two distinct propagation paths. As illustrated in Fig. 1, one component reaches the array via the direct line-of-sight path, whereas the other is a target-reflected replica of the transmitted waveform and experiences an additional propagation delay. As a result, the received data consist of two signal components impinging from distinct directions.

Assuming an array of M sensors and under the narrowband and far-field assumptions, the passively received complex

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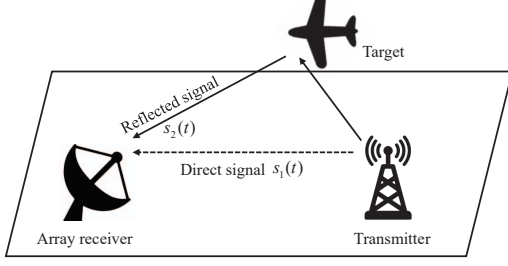


Fig. 1. Illustration of the PCL scenario.

signal is sampled and modeled as

$$\mathbf{y}[n] = \mathbf{A}(\boldsymbol{\theta})\mathbf{s}[n] + \mathbf{w}[n], \quad n = -\frac{N}{2}, \dots, \frac{N}{2} - 1, \quad (1)$$

where n denotes the snapshot index. The received signal vector $\mathbf{y}[n]$ represents the collection of sensor output samples at a specific time instance, and the noise vector $\mathbf{w}[n]$ is assumed to be zero-mean complex white Gaussian noise with covariance matrix $\sigma^2 \mathbf{I}_M$. The array manifold matrix is given by

$$\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1) \quad \mathbf{a}(\theta_2)], \quad (2)$$

where $\mathbf{a}(\theta_k)$ represents the steering vector corresponding to the k -th signal, with $k = 1$ denoting the direct signal and $k = 2$ denoting the reflected signal. The specific form of $\mathbf{a}(\theta_k)$ is determined by the array geometry. For instance, for a uniform linear array (ULA) with inter-element spacing d , the steering vector can be expressed as

$$\mathbf{a}(\theta_k) = \begin{bmatrix} 1 & e^{j\frac{2\pi d}{\lambda} \sin \theta_k} & \dots & e^{j\frac{2\pi d}{\lambda} (M-1) \sin \theta_k} \end{bmatrix} \quad (3)$$

where λ denotes the signal wavelength.

The signal vector $\mathbf{s}[n] = [s_1[n], s_2[n]]^T$ consists of two components derived from sampling the same continuous-time signal, each corresponding to a distinct propagation path. Specifically, $s_1[n]$ represents the direct signal, while $s_2[n]$ denotes the reflected signal. Relative to the direct signal, the reflected signal experiences amplitude attenuation, a phase shift, a time delay, and a Doppler-induced frequency shift. Accordingly, the two signals can be expressed as

$$\begin{aligned} s_1[n] &= s_1(T_s n), \\ s_2[n] &= \alpha e^{j\phi} s_1(T_s(n - \tau)) e^{j\omega_d n}, \end{aligned} \quad (4)$$

where $s_1(t)$ denotes the complex envelope of the continuous-time signal, T_s is the sampling interval, τ and ω_d represent the relative time delay and Doppler shift in the discrete-time domain, α and ϕ denote the unknown relative amplitude attenuation and phase shift of the reflected signal.

Let $\mathbf{Y} \in \mathbb{C}^{M \times N}$ denote the data matrix formed by collecting N snapshots, which can be written as

$$\mathbf{Y} = \mathbf{A}(\boldsymbol{\theta})\mathbf{S} + \mathbf{W}, \quad (5)$$

where each column of $\mathbf{Y}$ corresponds to the array observation at a given snapshot, and $\mathbf{W}$ represents additive noise. The signal matrix $\mathbf{S}$ contains two components associated with the direct-path signal and its reflected counterpart. In particular, the corresponding signal vectors are given by

$$\begin{aligned} \mathbf{s}_1 &= [s_1[-\frac{N}{2}] \quad \dots \quad s_1[\frac{N}{2} - 1]]^T, \\ \mathbf{s}_2 &= [s_2[-\frac{N}{2}] \quad \dots \quad s_2[\frac{N}{2} - 1]]^T. \end{aligned} \quad (6)$$

According to (4), the reflected signal is a delayed and Doppler-shifted version of the direct signal. When the relative delay τ is small compared with the total observation interval T , i.e., $|\tau| \ll T$, the linear time shift can be approximated by a circular shift. By exploiting the time-shift property of the discrete Fourier transform (DFT), the reflected signal vector can be expressed as a parametric transformation of the direct signal vector [17],

$$\mathbf{s}_2 \approx \alpha e^{j\phi} \mathbf{D}_w \mathbf{F}^H \mathbf{D}_\tau \mathbf{F} \mathbf{s}_1, \quad (7)$$

where $\mathbf{F}$ denotes the $N \times N$ DFT matrix, and $\mathbf{D}_\tau$ and $\mathbf{D}_w$ are diagonal matrices accounting for the time delay and Doppler shift, respectively. Specifically,

$$\mathbf{F} = \frac{1}{\sqrt{N}} \exp\left(-j\frac{2\pi}{N} \mathbf{n} \mathbf{n}^T\right), \quad (8)$$

and

$$\mathbf{D}_\tau = \text{diag}\{\exp(-j\frac{2\pi}{N} \tau \mathbf{n})\}, \quad \mathbf{D}_w = \text{diag}\{\exp(j\omega_d \mathbf{n})\}, \quad (9)$$

where $\mathbf{n} = [-\frac{N}{2}, \dots, \frac{N}{2} - 1]^T$.

Substituting (7) into (5), the overall observation model can be written as

$$\mathbf{Y} = \mathbf{A}(\boldsymbol{\theta}) \begin{bmatrix} \mathbf{s}_1^T \\ \alpha e^{j\phi} [\mathbf{Q}(\tau, \omega_d) \mathbf{s}_1]^T \end{bmatrix} + \mathbf{W}, \quad (10)$$

where $\mathbf{Q}(\tau, \omega_d) = \mathbf{D}_w \mathbf{F}^H \mathbf{D}_\tau \mathbf{F}$ compactly characterizes the joint effects of the propagation delay and Doppler shift. Therefore, the unknown parameter vector can be written as

$$\boldsymbol{\xi} = [\boldsymbol{\theta}^T \quad \boldsymbol{\beta}^T]^T. \quad (11)$$

where $\boldsymbol{\beta}$ represents all signal-related unknown parameters, given by

$$\boldsymbol{\beta} = [\alpha \quad \phi \quad \tau \quad \omega_d \quad \Re\{\mathbf{s}_1\}^T \quad \Im\{\mathbf{s}_1\}^T]^T, \quad (12)$$

where $\Re\{\mathbf{s}_1\}$ and $\Im\{\mathbf{s}_1\}$ denote the real and imaginary parts of the direct signal vector $\mathbf{s}_1$, respectively. Accordingly, the observation model can be summarized in the compact form

$$\mathbf{Y} = \mathbf{A}(\boldsymbol{\theta})\mathbf{S}(\boldsymbol{\beta}) + \mathbf{W}, \quad (13)$$

which serves as the basis for the estimation method considered in the following sections.

III. EM-BASED JOINT ESTIMATION METHOD

In this section, we focus on the model given in (13) and derive a joint estimation method within the ML framework using the EM algorithm.

A. Maximum Likelihood Estimation

ML estimation is the most fundamental estimation approach, renowned for its asymptotic efficiency and its essential role in solving complex estimation problems [20]. Under the assumption of Gaussian white noise, the ML estimator for the proposed model is formulated as the optimization problem:

$$\begin{aligned}\hat{\xi} &= \arg \max_{\xi} \ln p(\mathbf{Y}; \xi) \\ &= \arg \min_{\theta, \beta} \|\mathbf{Y} - \mathbf{A}(\theta)\mathbf{S}(\beta)\|_F^2,\end{aligned}\quad (14)$$

where ξ is the unknown vector defined in (11) and $\mathbf{Y}$ is defined in (13). The function $\ln p(\mathbf{Y}; \xi)$ represents the log-likelihood function of the studied model. The second line in (14) reformulates this problem using the Frobenius norm, providing a more compact representation of the optimization objective based on (13).

This optimization problem involves a high-dimensional search over all unknown parameters, including DOAs and signal parameters, resulting in a prohibitive computational complexity. To address this challenge, we employ the EM algorithm, which iteratively solves the optimization problem while significantly reducing computational complexity.

B. Expectation–Maximization Algorithm

EM algorithm is an iterative optimization technique specifically designed for ML estimation. It alternates between the expectation step (E-step), which computes the expected log-likelihood given the current parameter estimates, and the maximization step (M-step), which updates the parameter estimates to maximize the expected log-likelihood. This iterative approach provides an efficient update to solve the high-dimensional optimization problem while ensuring computational feasibility [15].

In the scenarios involving superimposed signals, the EM algorithm iteratively decomposes the observed data into its constituent signal components and refine the parameter estimates at each iteration [21]. Under the regularity conditions, the algorithm converges to a stationary point of the likelihood function.

In our model, the observed signal is a superposition of the direct and reflected signals. Specifically, the model in (1) can be reformulated as

$$\mathbf{y}[n] = \sum_{k=1}^2 \mathbf{s}_k[n; \xi] + \mathbf{w}[n], \quad (15)$$

with

$$\mathbf{s}_k[n; \xi] = \mathbf{a}(\theta_k)s_k[n], \quad k = 1, 2, \quad (16)$$

where $s_k[n; \xi]$ denotes the k -th signal component contributing to $\mathbf{y}[n]$, and the parameter vector ξ defined in (11) is explicitly introduced to reflect the dependence of the signals on the underlying parameters.

To apply the EM algorithm, we introduce a complete data vector $\mathbf{x}[n]$

$$\mathbf{x}[n] = [\mathbf{x}_1^T[n] \quad \mathbf{x}_2^T[n]]^T, \quad (17)$$

Algorithm 1 Joint Parameter Estimation via EM

Require: Observation matrix $\mathbf{Y}$, initial parameters $\xi^{(0)}$, maximum iterations $K_{\max}$, convergence threshold ϵ

Ensure: Estimated parameters $\hat{\xi}$

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1: for  $i = 0$  to  $K_{\max} - 1$  do
2:   E-step:
3:   Compute  $\mathbf{x}_k^{(i)}[n]$ ,  $k = 1, 2$ , using (20)
4:   M-step:
5:   Update  $\theta_k^{(i+1)}$ ,  $k = 1, 2$ , via (22)
6:   Reconstruct  $\hat{\mathbf{S}}^{(i+1)}$  using (23) and estimate
       $(\hat{\tau}^{(i+1)}, \hat{\omega}_d^{(i+1)})$  via (24)
7:   Update  $\xi^{(i+1)}$ 
8:   if  $\|\xi^{(i+1)} - \xi^{(i)}\| < \epsilon$  then
9:     break
10:  end if
11: end for
12: return  $\hat{\xi}$ 
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where

$$\mathbf{x}_k[n] = \mathbf{a}(\theta_k)s_k[n] + \mathbf{w}_k[n], \quad k = 1, 2. \quad (18)$$

The noise $\mathbf{w}_k[n] = \gamma_k \mathbf{w}[n]$ is assumed to be independent across different indices k , where each noise term is scaled by a factor γ_k subject to the constraint $\sum_{k=1}^2 \gamma_k^2 = 1$. The complete data and the observed data are related through a many-to-one mapping, which can be expressed as:

$$\mathbf{y}[n] = [\mathbf{I}_M \quad \mathbf{I}_M] \mathbf{x}[n] = \mathbf{H}\mathbf{x}[n]. \quad (19)$$

Following the derivation in [21], EM algorithm for our estimation problem is:

1) *E-Step:* For each k , compute the conditional expectation of the complete data, which effectively partitions the observed data $\mathbf{y}[n]$ into source-specific components. Specifically, we have

$$\begin{aligned}\mathbf{x}_k^{(i)}[n] &= \mathbf{a}(\theta_k^{(i)})s_k^{(i)}[n] \\ &+ \gamma_k^2 \left[\mathbf{y}[n] - \sum_{l=1}^2 \mathbf{a}(\theta_l^{(i)})s_l^{(i)}[n] \right],\end{aligned}\quad (20)$$

where the current estimate $s_k^{(i)}[n]$ is computed based on the current signal parameter estimates $\beta^{(i)}$ (for simplicity, this dependence is not explicitly shown).

2) *M-Step:* Update the parameter estimates by maximizing the expected log-likelihood by solving

$$\xi^{(i+1)} = \arg \min_{\theta, \beta} \sum_{k=1}^2 \sum_{n=-N/2}^{N/2-1} \left\| \mathbf{x}_k^{(i)}[n] - \mathbf{a}(\theta_k)s_k[n] \right\|^2. \quad (21)$$

This optimization problem can be interpreted as performing ML estimation separately for each source, where the source-specific data $\mathbf{x}_k^{(i)}[n]$ obtained from the E-step is treated as the observed data corresponding to the respective signal.

It is important to note that, in the M-step, since the parameter vector ξ is shared by both signals, we cannot perform a

parallel search for the optimal parameters for each signal as in [21]. Instead, one must search over a high-dimensional parameter space to identify the optimal parameters that minimize the cost function. Although this approach yields the greatest increase in likelihood, it is also possible to relax the strict requirement to simply ensure an increase. Hence, a suboptimal solution may be selected in this step, as implemented by the generalized EM (GEM) algorithm [14].

To approximate the solution of the M-step with reduced computational complexity, we adopt a sequential update strategy. Specifically, we first apply conventional beamforming (CBF) to estimate the DOA of each source component separated in the E-step

$$\theta_k^{(i+1)} = \arg \max_{\theta} \sum_{n=-N/2}^{N/2-1} \left| \mathbf{a}^H(\theta) \mathbf{x}_k^{(i)}[n] \right|^2, \quad k = 1, 2. \quad (22)$$

Let $\hat{\mathbf{A}}^{(i+1)}$ denote the array manifold matrix constructed from the updated DOA estimates $\theta^{(i+1)}$ obtained via (22). Given $\hat{\mathbf{A}}^{(i+1)}$, the source waveforms are then reconstructed by

$$\hat{\mathbf{S}}^{(i+1)} = \left(\hat{\mathbf{A}}^{(i+1)H} \hat{\mathbf{A}}^{(i+1)} \right)^{-1} \hat{\mathbf{A}}^{(i+1)H} \mathbf{Y}. \quad (23)$$

Denote by $\hat{\mathbf{s}}_1^{(i+1)}$ and $\hat{\mathbf{s}}_2^{(i+1)}$ the first and second rows of $\hat{\mathbf{S}}^{(i+1)}$, corresponding to the reconstructed direct and reflected signals, respectively. These two waveform estimates are subsequently used to estimate the time delay and Doppler shift by cross-ambiguity function (CAF) [22]

$$\hat{\tau}^{(i+1)}, \hat{\omega}_d^{(i+1)} = \arg \max_{\tau, \omega_d} \left| \hat{\mathbf{s}}_2^{(i+1)H} \left(\mathbf{Q}(\tau, \omega_d) \hat{\mathbf{s}}_1^{(i+1)} \right) \right|, \quad (24)$$

where $\mathbf{Q}(\tau, \omega_d)$ is defined in (10). This completes the $(i+1)$ -th iteration. Finally, the overall procedure of this algorithm is summarized in Algorithm 1.

IV. SIMULATIONS

We evaluate the proposed EM-based joint estimation algorithm for passive coherent localization. For comparison, we implement a sequential baseline: it first estimates the DOAs using a conventional DOA approach (for a fair comparison, we also adopt an EM-based DOA estimator [23]), then reconstructs the source waveforms via least squares, and finally applies the CAF [22] to the reconstructed waveforms to estimate the delay and Doppler shifts. In addition to the empirical RMSEs, we plot the corresponding Cramér–Rao bound (CRB) for the joint approach as a theoretical lower-bound reference.

A uniform linear array (ULA) with $M = 6$ sensors (half-wavelength spacing) is used. The noise is spatially/temporally white with variance $\sigma^2 = 1$, and the observation length is $N = 100$. The transmit waveform is a sample realization of an ARMA process obtained by filtering a white Gaussian noise sequence with a predefined filter. The direct and reflected signal components are appropriately scaled to achieve the desired signal-to-noise ratios (SNRs). The ESPRIT algorithm is then employed to obtain initial DOA estimates, which are

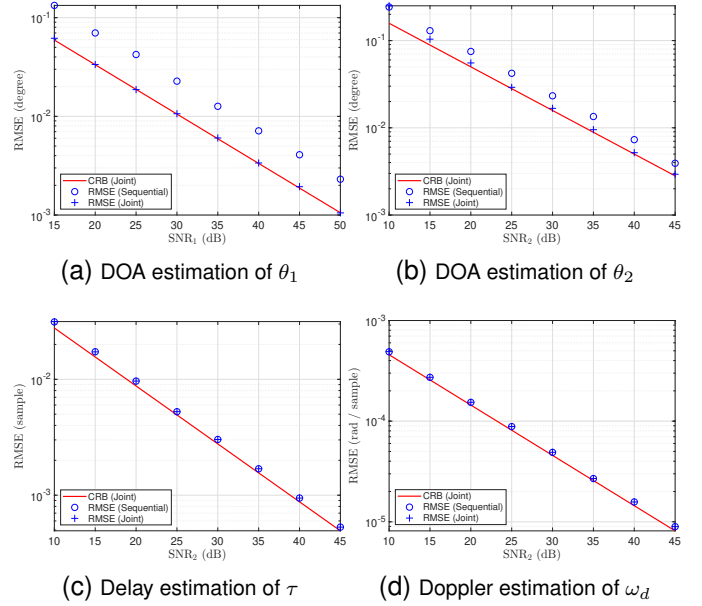


Fig. 2. RMSEs of the estimated parameters and the corresponding square roots of the CRBs versus SNR. Lines denote the square-root CRBs, while markers indicate the empirical RMSEs.

used to initialize both algorithms. All results are averaged over 1000 independent Monte Carlo runs.

We examine the estimation performance as a function of SNR. The DOAs are set to $\theta_1 = 0^\circ$ and $\theta_2 = 10^\circ$, with $\tau = 5$ samples and $\omega_d = 1 \times 10^{-4}$ rad/sample. The SNR gap is fixed at $\text{SNR}_1 - \text{SNR}_2 = 5$ dB, and SNR_1 (equivalently SNR_2) is swept across the tested range.

Fig. 2 shows the RMSEs of the four parameters versus SNR for both methods. As expected, the RMSEs decrease with increasing SNR for all parameters. The proposed EM-based joint approach consistently outperforms the sequential baseline. This gain mainly stems from the DOA estimation stage: unlike the sequential scheme, which treats the source waveforms as completely unknown, the joint method exploits the prior structural relationship between the sources—namely, that the two received signals are delayed replicas of the same waveform. For delay and Doppler estimation, however, Fig. 2 shows that the attainable accuracy of the joint and sequential methods is nearly identical. In fact, it can be shown theoretically that both approaches achieve the same performance bound for delay and Doppler under the assumed model. The simulation results therefore agree well with the theoretical analysis.

V. CONCLUSIONS

This paper proposed an EM-based maximum-likelihood method for joint estimation of DOA, delay, and Doppler in a PCL scenario with direct and reflected signals. Simulation results demonstrate improved DOA accuracy compared with sequential approaches, while maintaining comparable performance for delay and Doppler estimation, highlighting the benefit of leveraging signal-structure coupling in PCL.

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